

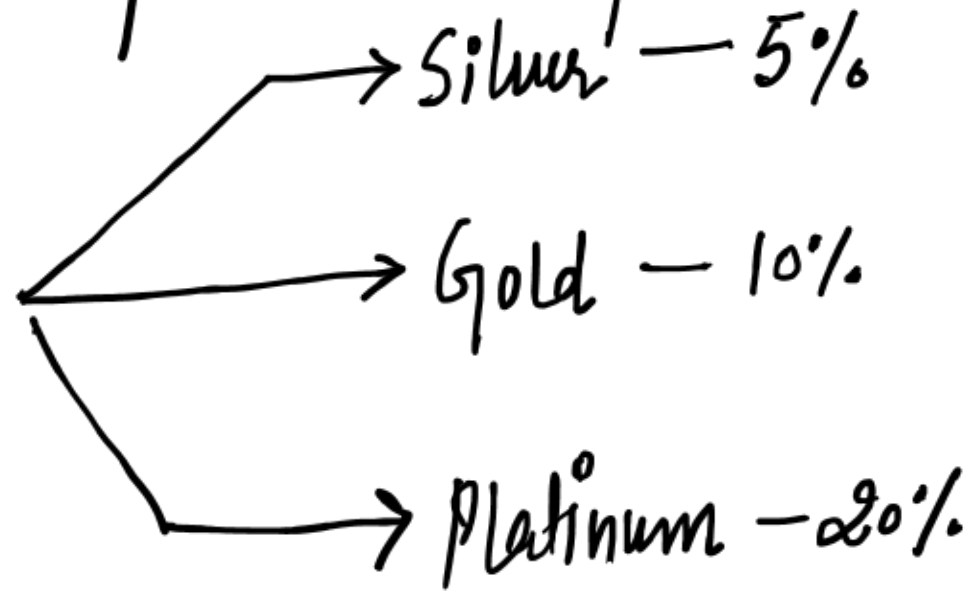
→ Clustering Algorithms (Unsupervised):

- * K-Means Clustering (imp) (Today's agenda)
- * Hierarchical clustering

→ Industrial Example:-

* Amazon → Great Indian festival sale

↓
Business team



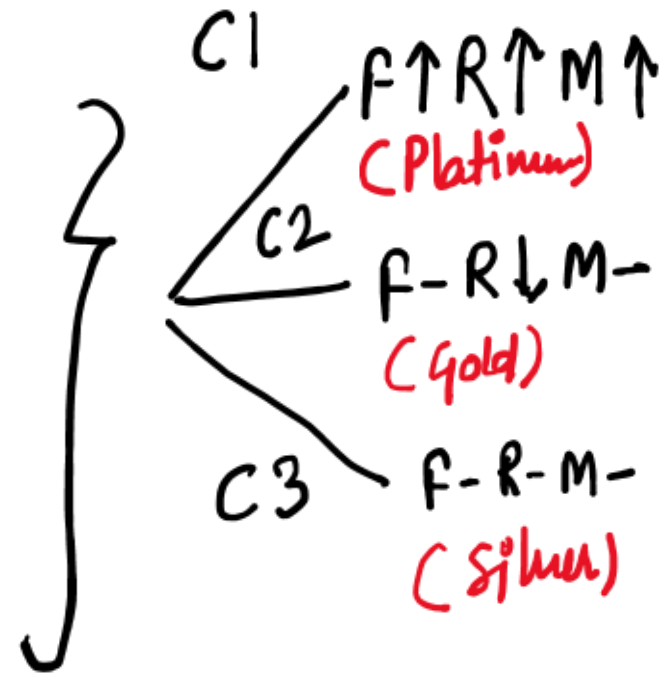
} Customers
[40M]

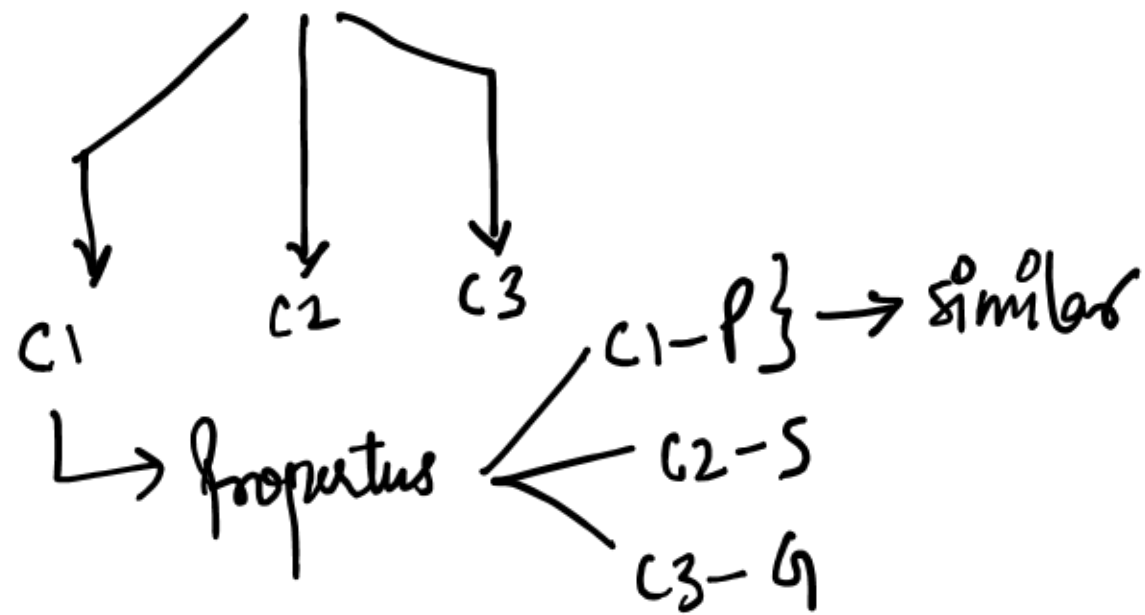
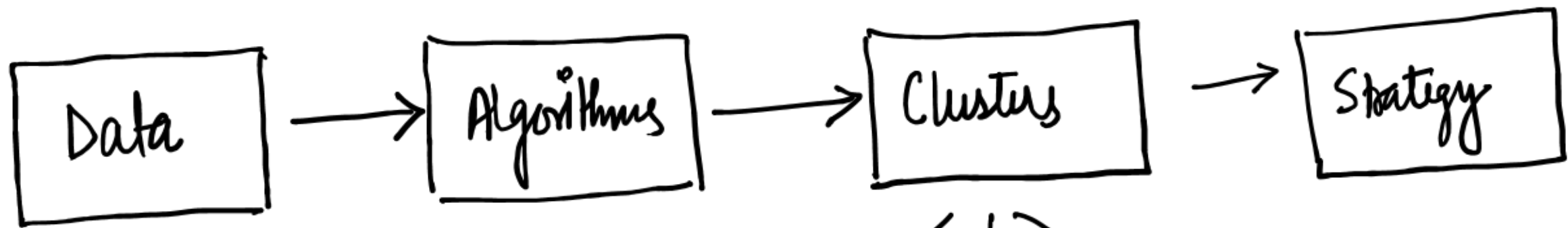
Strategy-1

- * Randomly distribute discount coupons among the customers

Strategy-2 (Better)

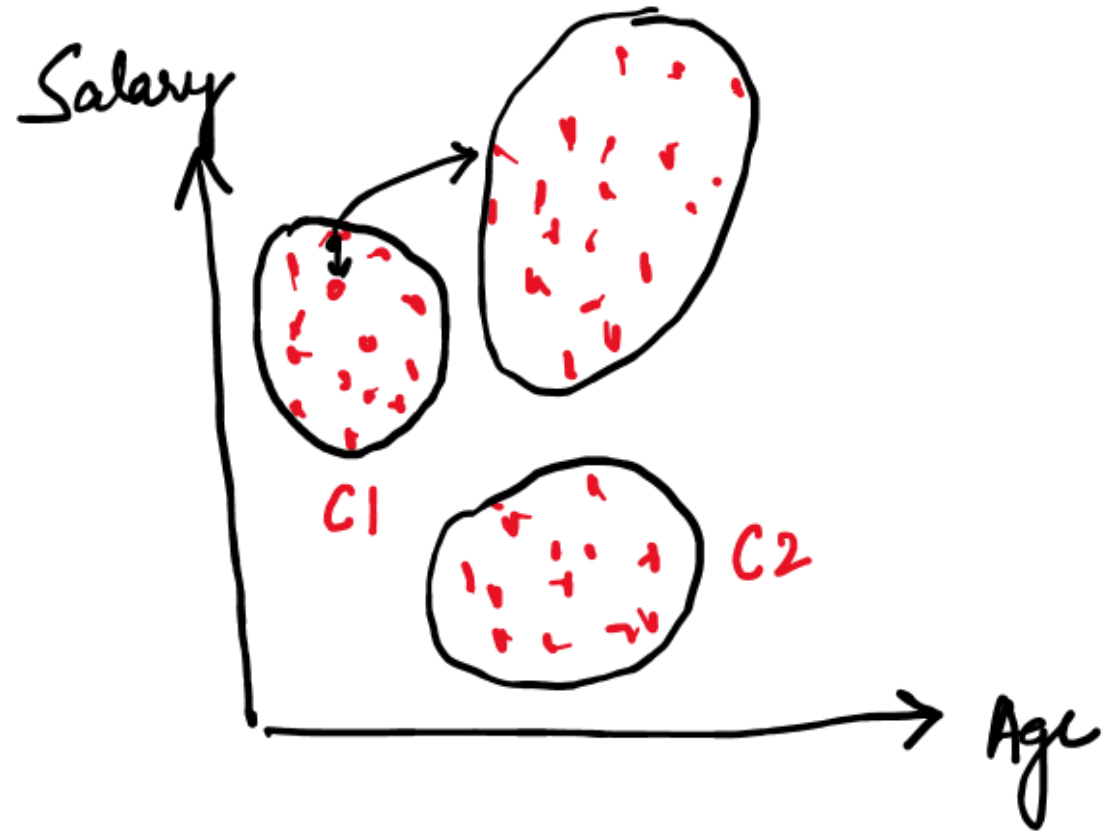
- * Analyse customer behaviour on platform
- * frequency
- * recently
- * Monetary





→ Density based Clustering :- (KNN, DBSCAN) ^{C3}

C1 → Age ↓ S ↑
C2 → Age (m) S ↓
C3 → Age ↑ S ↑



→ K-Means Clustering : (Unsupervised Algorithm) (Iterative)

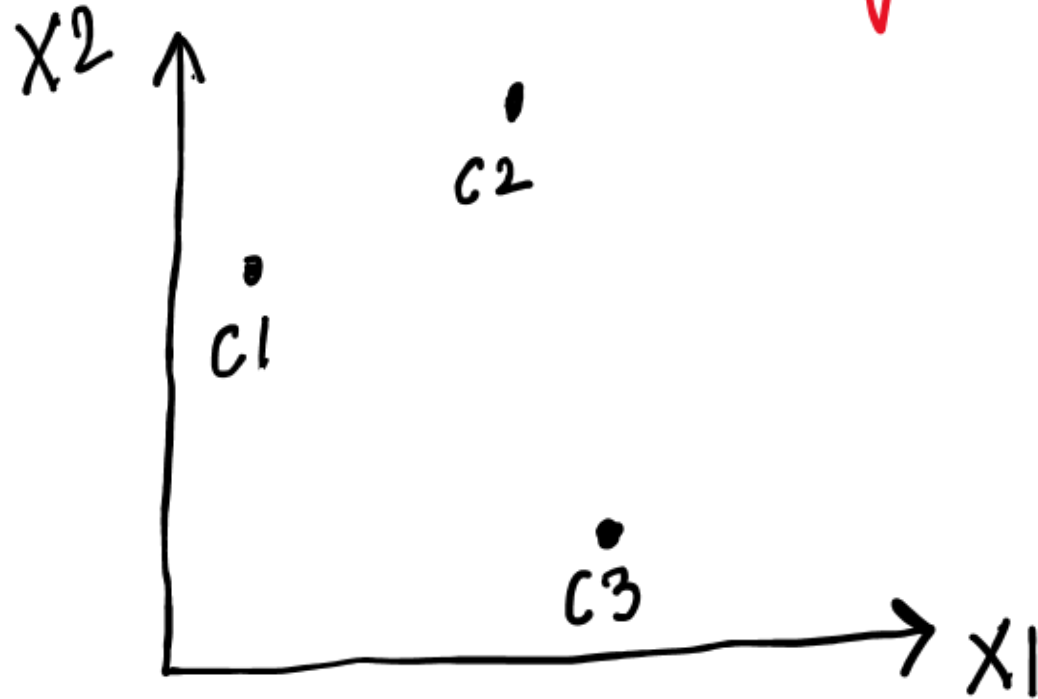
- * No. of clusters to be pre-known before clustering ✓
 - ↳ given by stakeholders (generally)
- * Applicable for only numerical columns
- * Scaling is mandatory here
- * Proximity measure
 - Euclidean (by default)
 - Manhattan
 - Minkowski distance

→ Algorithm :-

$K=3$

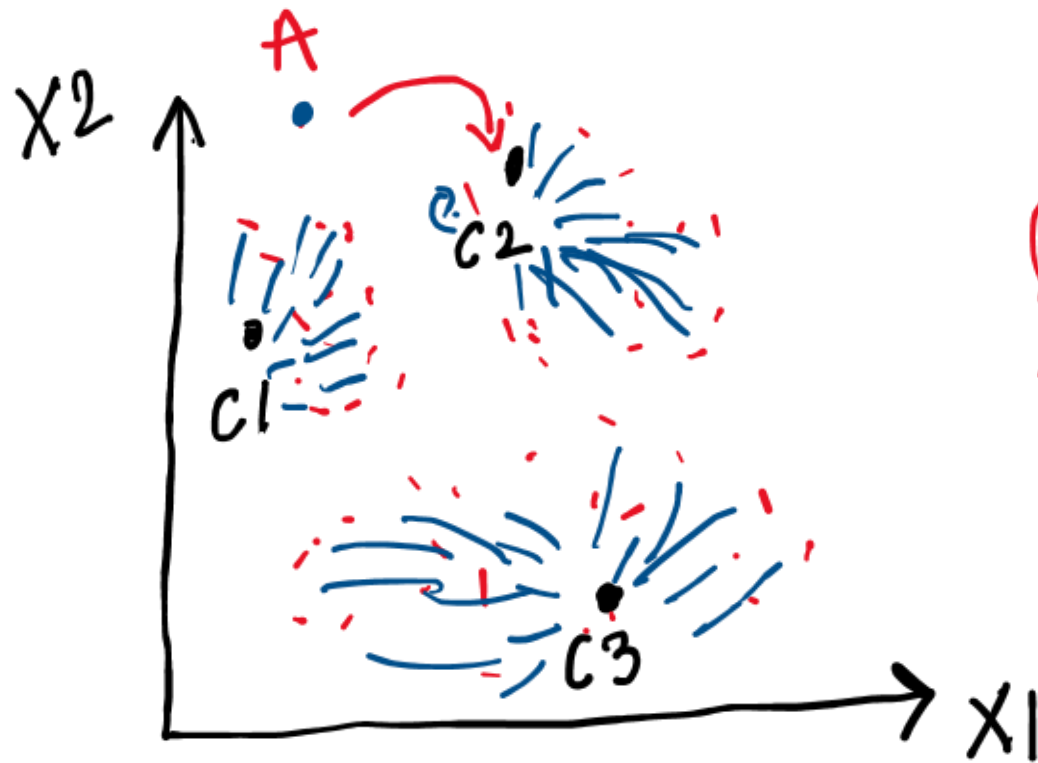
① Random Initialization :- Randomly initialise K centroids

↓
means



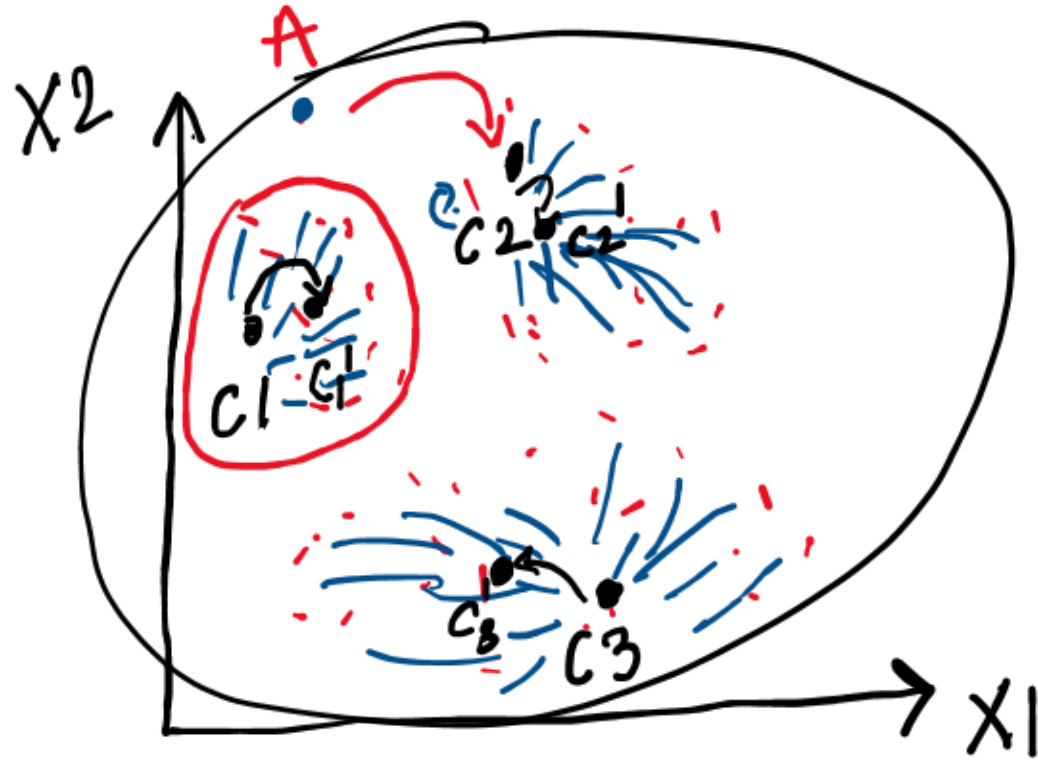
2- Assignment:- Calculate distance of every datapoint to each centroid, then that datapoint will be assigned to centroid which is closest.

$$d_2 < d_1 < d_3$$

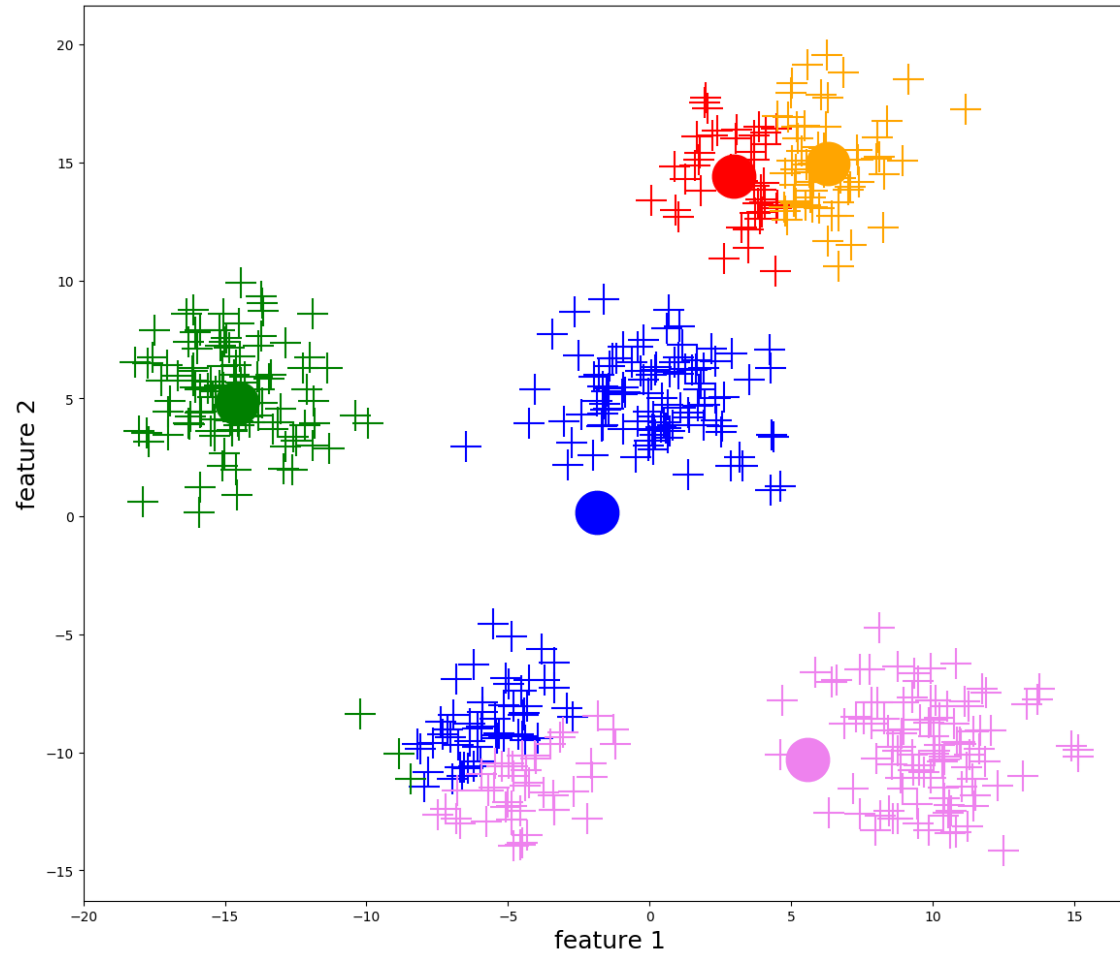


Process get repeated for every data point

3- Update :- Update all centroids with the mean of all datapoints assigned to it in previous step



4- Repeataation:- Repeat step-2 & step-3 until convergence
↓
Cluster stable

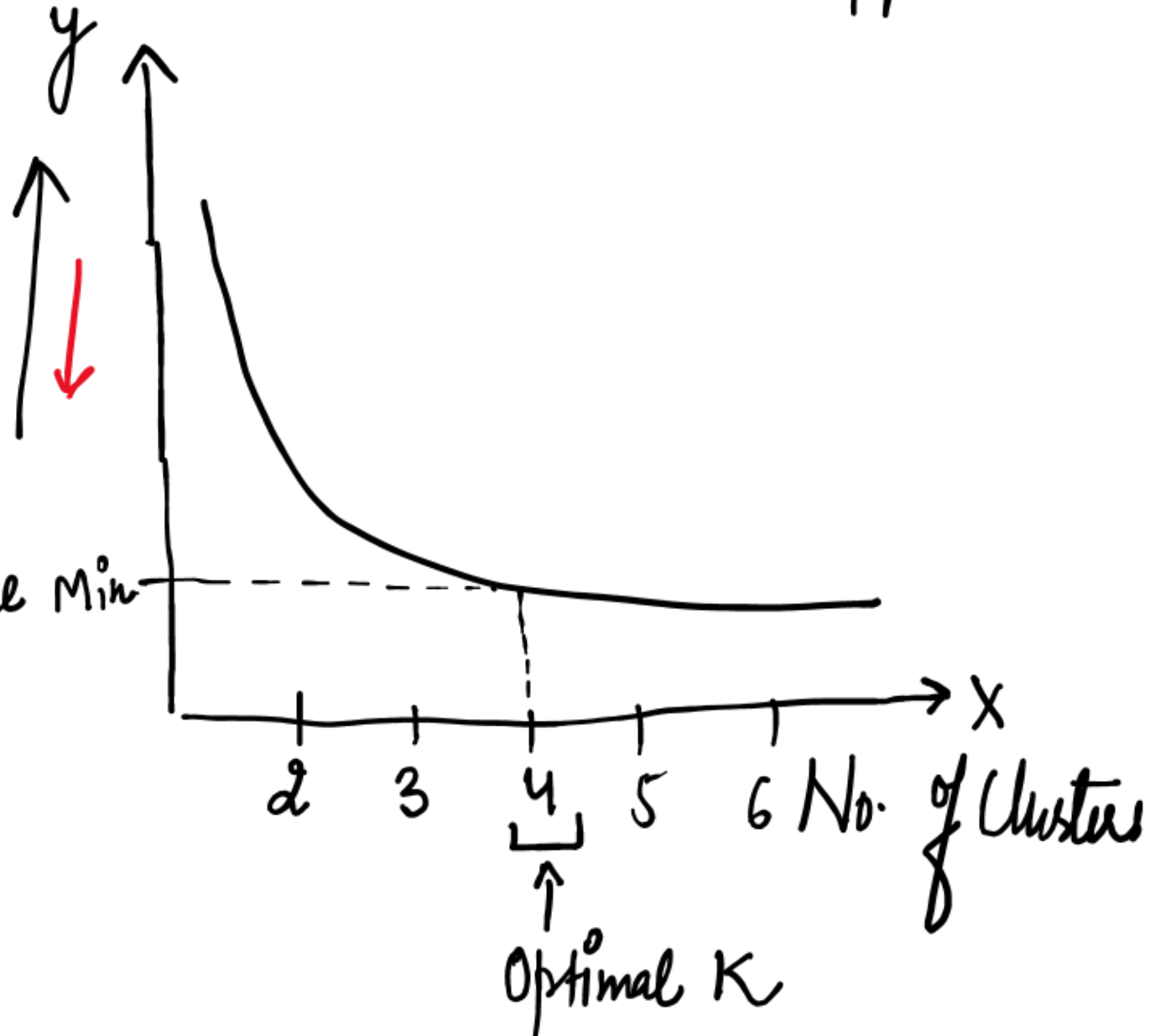


→ Optimal no. of clusters (K) :-

Sudden drop / Elbow

* Elbow method :-

WCSS → Error $\propto$



↓
Within cluster sum of square Min

→ Intra-cluster (↓)
{ Inter cluster (↑)

